VALUE AT RISK
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Project Report

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Abstract

We summary concepts of Value at Risk and some issues related to it.
Keyword: Value at Risk, Monte-Carlo, Backtestting ...

1 Introduction

During the past decade, Value at Risk (VaR) has become one of the standard measures of risk used by financial institutions and regulators. Conceptually, VaR measures the potential loss of a portfolio that will not be exceeded with a specified probability over a specified time horizon. Thus, VaR is merely the quantile of the distribution of a portfolio's future returns, conditional on any information available. Despite its conceptual simplicity, however, the measurement of VaR is by no means an easy statistical exercise, but a very challenging statistical problem.

2 Definition of Value at Risk

In finance, Value at risk is a **loss (level)** that we are fairly sure will **not be exceeded** with a **given probability** if the current portfolio is held **over some period of time**.
VaR has two basic parameters:

- the significance level α (or confidence level $1-\alpha$) – a given probability;
- the risk horizon , denoted h, which is the period of time, traditionally measured in trading days rather than calendar days, over which the VaR is measured.

The $100\alpha\%$ h-day VaR at time t is:

$$VaR_{ht,\alpha} = -x_{h,\alpha} \quad s.t \quad P(X_{ht} < x_{h,\alpha}) = \alpha$$

where X_{ht} denote h-day portfolio's return.
Mathematically, Value at Risk is actually a quantile of a random variable at a level of probability. And the mathematical problem is that you need to calculate this quantile without any assumption about distribution of the random variable but with a given data set of price series of related assets. You can consider a portfolio as an asset and treat them in the same way. That why you see here, we work primarily with only an asset. You can find some treatment with a portfolio at last.

3 Method for calculating VaR

3.1 Problem and notations

Given a daily price series of an asset:

$$S_1, S_2, ..., S_n$$

Loss is determined as

$$L_{t,t+h} = -(S_{t+h} - S_t)$$

Return of assets

$$R_{t,t+h} = \log\left(\frac{S_{t+h}}{S_t}\right) \approx \frac{S_{t+h} - S_t}{S_t}$$

Value at risk of the asset's return and loss are proportional:

$$P(L^p_{t,t+h} \geq VaR^L_{t,t+h}(\alpha)) = P\left(R^p_{t,t+h} \leq \frac{-VaR^L_{t,t+h}(\alpha)}{S_t}\right)$$

$$= P(-R^p_{t,t+h} \geq VaR^R_{t,t+h}(\alpha))$$

3.2 Scale time root:

In this paper we're going to compute daily VaR, however in practice, we also need to know weekly and month VaR. We need to obtain data of a sufficiently long period. The larger h is, the longer period of historical data is. So it can be lack of data. Fortunately, we can solve this problem by an acceptable transformation from daily to other time-scale VaR.
Let $x_{h,\alpha}$ denote the α quantile of h-day discounted log returns. We seek ξ such that

$$x_{h,\alpha} = \xi x_{1,\alpha}$$

In common, we use

$$\xi = \sqrt{h}$$

3.3 Different Approaches to VaR

3.3.1 Overview

To calculate VaR we need some assumptions. And each set of assumptions define a method. Typically, there three kinds of assumptions, and their corresponding methods:

***Historical method*:**

This is the most simple method. It assume the return at every time point is the same and consider them as observations of unique variable. And the solution is simply taking quantile of a data range. The method is also strengthen by using some adjustment to the historical data to make it present current level of price.

***Variance method (or theoretical method)*:**

This method is most complicated but also mathematically clearest. It make clear that VaR depends on volatility of the asset's return. So, we need some assumptions about distribution of the return to estimate its volatility.

***Simulation method*:**

This method is a corporation of two above methods. It use models to simulate asset price and then calculate VaR base on historical method.

3.3.2 Variance method

As we known, this method use models to calculate VaR. So, the most important thing is that the chosen model must be in accord with data. Of course, it is also the most difficult to find an appropriate model like that. Here, we need a model to estimate current volatility and forecast future volatility. So, we hope to go deeply this problem in other papers. Now we introduce some common models.

***Normal return*:**

In this model, we suppose that asset-returns are normal variables.

$$R^p_{t,t+h} \sim N\left(\mu_t, \sigma_t^2\right)$$

We have the following formula for VaR:

$$VaR^P_{t,t+h}(\alpha) = -\mu_t + \sigma_t \phi^{-1}(1-\alpha)$$

The extent problem is fitting data to a distribution. There are so many distributions we can choose, of couse, include compound distributions and mixed distributions. If normal distribution is not good enough, you can use mixed normal distribution.

***Time series model*: ARMA, ARCH & GARCH**

For example, let consider GARCH(1,1) model:

$$\begin{cases} R_{t,t+h} = \sigma_t \epsilon_t \\ \sigma_t^2 = w + vR^2_{t-h,t} + \gamma\sigma^2_{t-h} \end{cases}$$

Here, we have

$$R_{t,t+h}|R^2_{t-h,t} \sim N(0, \sigma_t^2)$$

and

$$VaR_{t,t+h}(\alpha) = \sigma_t \phi^{-1}(1-\alpha)$$

One advantage is that, with time series models, we can estimate and forecast volatility at each specific time point, in past or in future.

3.3.3 Historical VaR:

This is the most common and easiest way to estimate VaR. In the historical simulation (HS) model, we'll use a large quantity of historical data to estimate VaR but makes minimal assumptions about the risk factor return distribution. That's why in some case, practitioners call this approach "Non-parametric VaR estimation". But remember that not using any assumption about parameters or distribution of asset returns doesn't mean we don't have any assumption on this approach. HS also divided into many ways to estimate the VaR level, such as:

***Basic Historical Simulation (HS)*:**

This is the most simple historical method. We only need the historical data of asset returns and then sort them from smallest to highest in term of value. After sorting, we can compute a cumulative distribution and plot the histogram. Finally we apply the quantile to calculate the VaR level.

***Exponential Weighted Historical Simulation (EW-HS)*:**

One of weakness of basic HS is the requirement of many historical data. That may lead the VaR to be unfitted to the status quo. So that we need to adjust the probability of each daily return to make the VaR level more update.

- **Step 1:** First we fix a smoothing constant, denoted λ as usual, between 0 and 1.
- **Step 2:** Then assign the probability weight $1-\lambda$ to the most recent observation on the return X_T, $T+1$ is the day we want to compute VaR.
- **Step 3:** Continue assign the weight $\lambda^h(1-\lambda)$ to the return X_{T-h}.When the weights are assigned in this way, the sum of the weights is 1, i.e. they are probability weights.
- **Step 4:** We use these probability weights to find the cumulative probability associated with the returns when they are put in increasing order of magnitude.

It is a hard problem to find an appropriate value of λ. It shouldn't too large, cause cumulative probability that equals to $1-\lambda^T$, too small. It also shouldn't too small, cause probabilities of returns quickly decrease to 0.

***Volatility Adjusted Historical Simulation (VA-HS)*:**

The weakness of both ways above is the change of volatility by the time. In 2 ways above, we just concern about the cumulative distribution of returns. However, trading on the stock market looks like sailing on the sea; before the storm, the sky is very nice but when the storm came everything is different. On stock market, your portfolio may volatile not too much in normal day, but suddenly "crash", the volatility will very high but VaR estimation by HS will be diluted by too many normal in data. So that, we need a volatility-adjusted way to compute VaR.

This method was suggested by Duffie and Pan (1997) and Hull and White (1998). It assume that the volatility of return change by time, denoted $t = 1, \ldots, T$. This is closer to real world than assumption that volatility of return is invariance. So we need estimate the volatility of the return $\{\hat{\sigma}_t\}_{t=1}^{T}$. This method is designed to weight returns in such a way that we adjust their volatility to the current volatility.We consider two ways to implement volatility adjustment :

- **EWMA model:**

$$\hat{\sigma}_t = (1-\lambda) r_{t-1}^2 + \lambda \hat{\sigma}_{t-1}^2, \quad t = 2, ..., T$$

  where $\{r_t\}$ is log return,λ denote smoothing constant, $0 < \lambda < 1$ and $\hat{\sigma}_1^2$ may be set arbitrarily, or equal to r_1^2 .

- **GARCH (1,1):**

$$\hat{\sigma}_{t+1} = \hat{\omega} + \hat{\alpha} r_t^2 + \hat{\beta} \hat{\sigma}_t$$

Then we adjust the volatility to the return:

$$\tilde{r}_{t,T} = \left(\frac{\hat{\sigma}_T}{\hat{\sigma}_t} \right) r_t$$

***Kernel Fitting (KF-HS)*:**

The previous technique is effective if we use a very large sample and measuring quantiles no more extreme than 1%. How if we compute historical VaR at very extreme quantiles or if we can not obtain a very large sample?

Kernel Fitting can solve this problem. It fit a continuous distribution to the historical distribution of volatility adjusted return.
For example, Gaussian kernel:

$$K(x) = (2\pi)^{-1/2} e^{-x^2/2}$$

And the kernel density estimator is defined as:

$$\hat{f}(x) = \frac{1}{n} \sum_{i=1}^{n} \frac{1}{h} K\left(\frac{x - X_i}{h}\right)$$

where $X_1, ..., X_n$ denote the historical return, h denote the chosen bandwidth.

3.3.4 Monte Carlo VaR:

Computing VaR with Monte Carlo Simulations follows a similar algorithm to the one we used for Historical Simulations. In this method, we use a model to simulate data, then calculate VaR from that data. The method is executed with the algorithm with steps as below [RB]:

For each asset in the portfolio, we simulate the path of stock price.

- **Step 1:** Determine the length T of the analysis horizon and divide it equally into a large number N of small time increments $\Delta t = \frac{T}{N}$. We often choose $\Delta t = 1$.

- **Step 2:** Draw a random number from a random number generator and update the price of the asset at the end of first time increment:
  We use the standard stock price model to simulate the path of a stock price from the i^{th} day as defined by:

  $$S_{i+1} = S_i + S_i * (\mu \delta t + \sigma \phi \delta t^{1/2})$$

  where

  R_i is the return of the stock on the i^{th} day,
  S_i is the stock price on the i^{th} day,
  μ is the sample mean of the stock return,
  δt is the time step,
  σ is sample volatility(standard deviation) of the stock return,
  ϕ is a random number generated from a standard normal distribution.

At the end of this step/day ($\delta t = 1$ day), we have drawn a random number and determined S_{i+1} since all other parameter can be determined or estimated.

- **Step 3:** Repeat Step 2 until reaching the end of the analysis horizon T by walking along the N time intervals:
  At the next step/day ($\delta t = 2$), we draw another random number and apply as Step 2 to determine S_{i+2} from S_{i+1} . We repeat this procedure until we reach T and can determine S_{i+T} .

- **Step 4:** Repeat Step 2 and 3 a large number M of times to generate M different paths for the stock over T . It is an industry standard to run at least $10,000$ simulations even if $1,000$ simulations provide an efficient estimator of the terminal price of most assets.

- **Step 5:** Sort the M terminal stock prices from the smallest to largest, read the simulated value in this series that corresponds to the desired $(1-\alpha)\%$ confidence level (95% or 99% generally) and deduce the relevant VaR, which is the difference between S_i and the α-th lowest terminal stock price:
  After sorting we have:

$$S_{i+T}^{(1)} < S_{i+T}^{(2)} < ... < S_{i+T}^{(M)}$$

  Caculate VaR:

$$VaR(\alpha) = S_{i+T}^{(k)} - S_i, \quad k - 1 \leq M(1-\alpha) \leq k.$$

  .

3.4 Price Limit

The estimation of VaR become more difficult for emerging markets because trading in those markets is typically subject to varying degrees of regulatory restrictions and market imperfections. In [PH], the authors focus on the problems resulting from two most common and important forms of regulatory constraints and market imperfection, namely price limits and infrequent trading. Without considering micro structure factors such as infrequent trading and regulations like price limits, the traditional estimates of distribution parameters may be biased, thereby resulting in biased estimates of VaR measures. In Viet Nam Stock Market, there is only price limit problem, so we consider this problem. We use two limits Tobit model to describe the problem:

$$r = \begin{cases} L & if L > r^* \\ r^* & if L \leq r^* \leq U \\ U & if r^* > U \end{cases}$$

Where r^* is latent return, r is observed return and L, U are limits.

***Variance-Covariance method*:**

Intuitively, the usual estimates of volatility is biased downward, which results in a downward bias in the VaR estimate and a higher failure rate. For example, normal return model is used.
We wouldn't use model for r but for r^*. That mean

$$r^* \sim N(\mu, \sigma)$$

In a extensional way, we also can use regression equation

$$r_t^* = \alpha + \beta r_{mt} + \epsilon$$

Where $\epsilon \sim N(0, \sigma)$ and r_{mt} is a variance-known variable. And the parameters are deduced by Maximum Likelihood Estimator (MLE).

***Historical method*:**

In this method, if we use the simple empirical quantile for VaR then no repair is needed. However, if we need estimate parameters for adjust data, we must correct these estimations. Consider volatility adjust by Garch(1,1). We introduce Garch(1,1) with latent variable:

$$r_{it}^* = \sigma_{it|t-1}\epsilon_t, \quad \epsilon_t \sim N(0,1)$$

$$\sigma_{it|t-1}^2 = \omega + \alpha_i \sigma_{it-1|t-2} + \beta_i r_{it-1}^{*2}$$

And we also estimate parameters by MLE. After then, we adjust returns data by using follow formula:

$$\tilde{r}_{it} = \frac{\sigma_{iT+1}}{\sigma_{it}} r_{it}, \quad t = 1, ..., T$$

***Monte-Carlo method*:**

When apply this method, we only need add limit to simulated process. We note that the efficient of this method depend on parameters. So, we also need to estmate parameters using Tobit model.

4 Back-tests:

4.1 Overview

As same as other method in finance, VaR also need to be back-tested. We can have many different approaches to compute VaR, but the problem we need to solve is which approach has the best performance in practice.

Back-test data

The major problem is which data set is use to back-test: in-sample, out-sample or simulation. We call it In-sample, when we use exactly data that used to compute VaR. In this case, the basic historical method, hence, is the best. If we use simulation data, then theoretical methods and simulation method that based on the same model are best methods. In real, it is logical to use out-sample or future data to test VaR. If nothing change in the trend of data, so basic historical method is still useful. In other situation, every methods face challenge and only methods that use appropriate model can stably stand.

Null hypothesis

In back-test we want frequency of loss over VaR is exactly α. Actually, it is rare to get exactly that frequency and just a closed ratio can be accepted. Theoretically, the null hypothesis $H_0 : p = \alpha$ can be verified with a significant level. We denote p for the probability of loss cross over VaR. From observation of frequency, that equals to number of outstanding values over number of data points, we need to decide to accepted or reject the null hypothesis. This deciding tell us estimated VaR is good or bad.

p-Value

p-Value is computed from frequency (or observation) and compared to significant level to figure out the null hypothesis is accepted or rejected. In general, we define p-Value as probability of observing that frequency under assumption that null hypothesis is true.
Denoting the number of exceptions as x and the total number of observations as T, we may define the failure rate as x/T. In an ideal situation, this rate would reflect the selected confidence level. For instance, if a confidence level of 99% is used, we have a null hypothesis that the frequency of tail losses is equal to $p = (1–c) = 1–0.99 = 1\%$. Assuming that the model is accurate, the observed failure rate x/T should act as an

unbiased measure of p, and thus converge to 1% as sample size T is increased.
Each trading outcome either produces a VaR violation exception or not. This sequence of successes and failures is commonly known as Bernoulli trial. The number of exceptions x follows a binomial probability distribution:

$$f(x) = \binom{T}{x} p^x (1-p)^{T-x}$$

As the number of observations increase, the binomial distribution can be approximated with a normal distribution:

$$Z = \frac{x - pT}{\sqrt{p(1-p)T}} \approx N(0,1)$$

where pT is the expected number of exceptions and $p(1–p)T$ the variance of exceptions.
Visually, we think that $f(x)$ is maximum at $x = pT$. That is right and we can easy prove it. Denote

$$R(x) = \frac{f(x)}{f(x-1)} = \frac{T-x+1}{x} * \frac{p}{1-p}$$

We see that $R(x)$ decreasing and the last point at which $R(x) > 1$ is $x_0 = [(T+1)*p]$. That mean, $f(x)$ increase to $f(x_0)$ and decrease after that. Now we define

$$p - value = Prob\left(|X - x_0| \geq |x - x_0|\right)$$

If $p - value < 0.05$ we reject H_0, and in contrary, we accept H_0. This is a test at 0.05 significant level.
So,back-testing is a statistical problem where actual losses is compared to corresponding VaR estimates. When we have $p = 1\%$, it means that only 1 time-step in 100 time-steps occur an exception. This kind of back-testing called “unconditional coverage”. However a good VaR model not only compute correct amount of exceptions but also exceptions that are evenly spread over time i.e. are independent of each other. Clustering of exceptions indicates that the model does not accurately capture the changes in market volatility and correlations. Tests of conditional coverage therefore examine also conditioning, or time variation, in the data. Now we’ll introduce you very basic idea of each test.

4.2 Unconditional vs conditional Coverage Test

4.2.1 Unconditional Coverage

The most common test of a VaR model is to count the number of VaR exceptions in a time period (usually a day) when portfolio losses exceed VaR estimates. If the number of exceptions is less than α would indicate overestimation of risk. On the contrary, too many exceptions signal underestimation of risk. As we know, we rarely observe the exact amount of exceptions suggested by α. Instead of that, we'll study whether the number of exceptions is reasonable or not, i.e. will the model be accepted or rejected.

Kupiec POF-Test

The most widely known test based of failure rates has been suggested by Kupiec (1995). Kupiec's test, also known as the POF-test (proportion of failures), measures whether the number of exceptions is consistent with the confidence level.

The null hypothesis for the POF-test is: $H_0 : p = \hat{p} = \frac{x}{T}$ The idea is to find out whether the observed failure rate $\hat{p}$ is significantly different from p, the failure rate suggested by the confidence level. According to Kupiec (1995), the POF-test is best conducted as a likelihood-ratio (LR) test. The test statistic takes the form

$$LR_{POF} = -2 \ln \ln \left(\frac{(1-p)^{T-x} p^x}{\left[1 - \left(\frac{x}{T}\right)\right]^{T-x} \left(\frac{x}{T}\right)^x} \right)$$

Under the null hypothesis that the model is correct, LRPOF is asymptotically χ^2 (chisquared) distributed with one degree of freedom. If the value of the LRPOF –statistic exceeds the critical value of the χ^2 distribution, the null hypothesis will be rejected and the model is inaccurate. According to Dowd (2006), the confidence level for any test should be selected to balance between type 1 and type 2 errors. 95% is a common level because it implies that the model will be rejected only if the evidence against it is fairly strong.

Kupiec's POF-test has 2 weakness. First, the test is statistically weak with sample sizes consistent with current regulatory framework (one year) and has already been recognized by Kupiec himself. Secondly, POF-test considers only the frequency of losses and not the time when they occur. As a result, it may fail to reject a model that has clustered exceptions. Thus, model backtesting should not rely solely on tests of unconditional coverage. (Campbell, 2005)

Tuff test

Kupiec (1995) has also suggested another type of backtest, namely the TUFF-test (time until first failure). This test measures the time (ν) it takes for the first exception to occur and it is based on similar assumptions as the POF-test. The test statistic is a likelihood-ratio:

$$LR_{TUFF} = -2 \ln \ln \left(\frac{(p(1-p)^{\nu-1}}{\left(\frac{1}{\nu}\right)\left(1-\frac{1}{\nu}\right)^{\nu-1}} \right)$$

Here again, LRTUFF is distributed as a χ^2 with one degree of freedom. If the test statistic falls below the critical value the model is accepted, and if not, the model is rejected. The problem with the TUFF-test is that the test has low power in identifying bad VaR models. For example, if we calculate daily VaR estimates at 99% confidence level and observe an exception already on day 7, the model is still not rejected. (Dowd, 1998). Due to the severe lack of power, there is hardly any reason to use TUFF-test in model backtesting especially when there are more powerful methods available. As Dowd (1998), the TUFF-test is best used only as a preliminary to the POF-test when there is no larger set of data available. The test also provides a useful framework for testing independence of exceptions in the mixed Kupiec-test by Haas (2001).

4.2.2 Conditional Coverage

The unconditional coverage tests focus only on the number of exceptions. However, in term of risk management, we must check that whether clustering of exceptions or not. A clustering of exceptions means that you may lose too much in very near time points, hence a crash may happen. Good VaR models are capable of reacting to changing volatility and correlations in a way that exceptions occur independently of each other, whereas bad models tend to produce a sequence of consecutive exceptions. (Finger, 2005). So that tests of conditional coverage try to deal with this problem by not only examining the frequency of VaR violations but also the time when they occur. We'll present two conditional coverage tests: Christoffersen's interval forecast test and the mixed Kupiec-test by Haas (2001).

Mixed Kupiec-Test

Haas (2001) introduces an improved test for independence and coverage, using the ideas by Christoffersen and Kupiec. Haas proposes a mixed Kupiec-test which measures the time between exceptions instead of observing only whether an exception today depends on the outcome of the previous day.

Thus, the test is potentially able to capture more general forms of dependence.

According to Haas (2001), the Kupiec's TUFF-test, which measures the time until the first exception, can be utilized to gauge the time between two exceptions. The test statistic for each exception takes the form

$$LR_i = -2 \ln \ln \left(\frac{(p(1-p)^{\nu_i - 1}}{\left(\frac{1}{\nu_i}\right)\left(1 - \frac{1}{\nu_i}\right)^{\nu_i - 1}} \right)$$

where ν is the time between exceptions i and $i-1$.For the first exception the test statistic is computed as a normal TUFF-test. Having calculated the LR-statistics for each exception, we receive a test for independence where the null hypothesis is that the exceptions are independent from each other. With n exceptions, the test statistic for independence is

$$LR_{ind} = \sum_{i=1}^{n} LR_i$$

which is a χ^2 distributed with n degrees of freedom. The independence test can be combined with the POF-test to obtain a mixed test for independence and coverage, namely the mixed Kupiec-test:

$$LR_{mix} = LR_{POF} + LR_{ind}$$

TheLR_{mix}-statistic is χ^2 distributed with n+1 degrees of freedom. Just like with other likelihood-ratio tests, the statistic is compared to the critical values of χ^2 distribution. If the test statistic is lower, the model is accepted, and if not, the model is rejected.

4.3 VaR Strategy Line

In financial practice, one should review and modify reserve for risk frequently. Because risks are usually unstable, they change daily, so we must use as much as possible available information to calibrate them. With companies who are using VaR as a reserve measure, they may recalculate VaR immediately after they have new observation of their own asset value. These VaRs are usually updated daily and over time we have VaR lines as figure 1

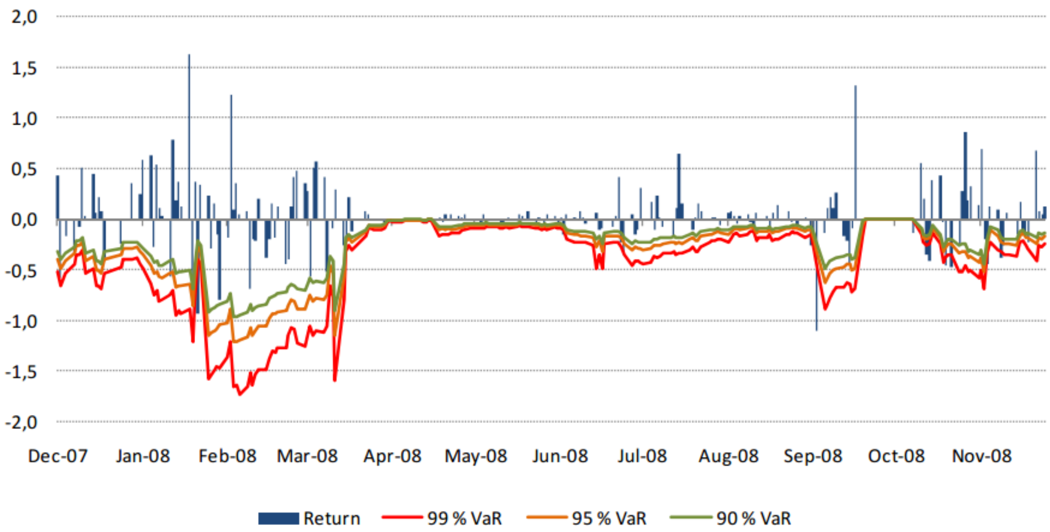

Figure 1: VaR Strategy Example

These VaR lines show how the reserve should be changed together with the volatility of the asset price. It help to save money in periods that the volatility is low and make sure the risk is backup safely in periods that the volatility is high. VaR lines are different depend on which method is use to compute VaR and then they show how efficient that method is.

5 Application

5.1 Tool

To demonstrate computing methods, we built a R package, named jvnVaR. Every one can download it from CRAN and using its functions. Now, let see what we can do. Firstly, you can use R help to know about the package. Or you can using the command:

```
> jListFunctions()
    [1] "jReturn : Compute return."
    [2] "jVaR : Compute Value at Risk."
    [3] "jVaRLim : Compute Value at Risk under limited price condition."
    [4] "jTestVaR : Test hypothesis of VaR."
    [5] "jMCPri : Simulate a series of price."
    [6] "jMCPriLim : Simulate a series of price under limited price condition."
```

[7] ”jStockList: List of stocks in Vietnam market.”
[8] ”jPrice : Historical price of a stock in Vietnam market.”

The function gives you a list of useful functions in the package. Let try some of them. You may want to use function ”jStockList” that give you a list of stocks on Vietnam Stock Market.

```
> ll <- jStockList()
> head(ll)
    [1] ”AAA” ”AAM” ”ABT” ”ACB” ”ACC” ”ACL”
```

You can get a price series of one of these stocks, using:

```
> s <- jPrice(’AAA’)
> head(s)
    [1] 48.7 46.5 46.3 45.2 45.1 45.1
```

The unit is one thousands of Vietnam Dong. After that, you can get its return series:

```
> r <- jReturn(s)
> head(r)
    [1] 0.047311828 0.004319654 0.024336283 0.002217295 0.000000000 -0.019565217
```

You get a return series and now may want to use R help to know about function ”jVaR”:

```
> help(jVaR)
```

Here is one example:

```
> VaR <- jVaR(’non_adjust_hist’, r , 0.05 , 0)
> VaR
    [1] 0.05405405
```

You got the VaR of the stock at a probability level of 0.05 and at last day in the series using basic historical method. If you want VaR at the previous (or next) day, replace 0 by -1 (or 1). Notice that there only several methods give different values between these days.
Now, you can back test this VaR. Assume we use the return series to back test:

> jTestVaR(r, VaR, 0.05, 0.05, 'p_value')
[1] 0.9377026 0.0500000 1.0000000

The first value of three above is p-value. The second value is the significant level of 0.05 that we put in 4th parameter. The last value, that equals 1 when p-value larger than significant level and equals 0 when other, let us know the VaR is accepted or rejected. You also try other tests:

> jTestVaR(r, VaR, 0.05, 0.05, 'pof')
[1] 0.000000 3.841459 1.000000

The first value of three above is POF statistic. The second value is the Chi-square quantile at the significant level of 0.05. The last value, that equals 1 when statistic smaller than Chi-square quantile and equals 0 when other.

> jTestVaR(r, VaR, 0.05, 0.05, 'tuff')
[1] 0.4130844 3.8414588 1.0000000

> jTestVaR(r, VaR, 0.05, 0.05, 'mixkup')
[1] 91.52520 60.48089 0.00000

We see that all methods accept VaR except mixed Kupiec test. That because only mixed Kupiec test using time of events that loss cross VaR.

5.2 Experiment

Table 1 show all results from one experience.
The result of rejection at mixed Kupiec test show that the sample is not a appropriate test data set.

5.3 VaR Strategy Compare

The below figures show the difference between the methods used to calculate VaR

VaR and In-Sample Backtest with 95% confidence level

Methods	VaR	Mixed Kupiec	POF	TUFF	p_Value
Basic HS	0.05405405	Reject	Accept	Accept	Accept
EW-HS	0.06165736	Reject	Reject	Accept	Reject
EWMA-HS	0.02839331	Reject	Reject	Accept	Reject
GARCH-HS	0.04364503	Reject	Reject	Accept	Reject
KERNEL-HS	0.07864407	Reject	Reject	Accept	Reject
EWMA-KERNEL	0.05739019	Reject	Accept	Accept	Accept
GARCH-KERNEL	0.07335799	Reject	Reject	Accept	Reject
GARCH11	0.04556064	Reject	Reject	Accept	Reject
NORMAL	0.04990335	Reject	Accept	Accept	Accept
SIMULATION	0.05088335	Reject	Accept	Accept	Accept
LIMIT(-0.1,0.1)	-	-	-	-	-
GARCH11	0.04558477	Reject	Reject	Accept	Reject
GARCH-HS	0.04368043	Reject	Reject	Accept	Reject
SIMULATION	0.04891968	Reject	Reject	Accept	Reject

Table 1: Test result

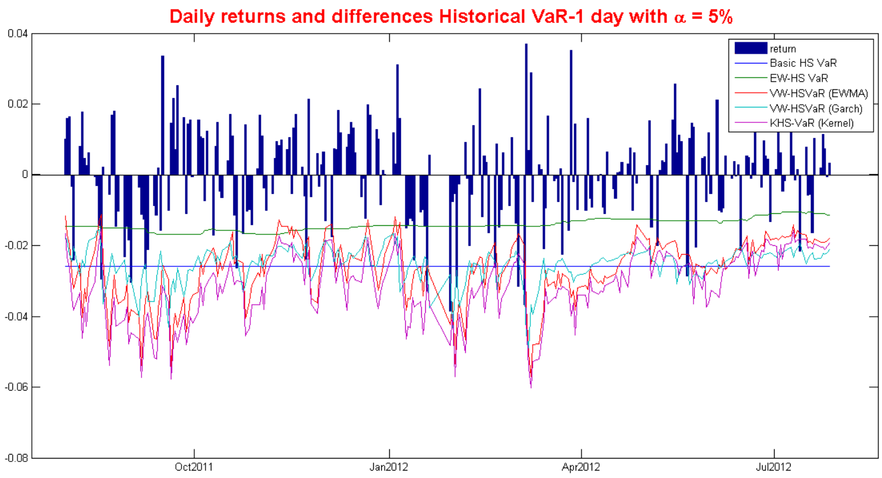

Figure 2: VaR Strategies Difference 1

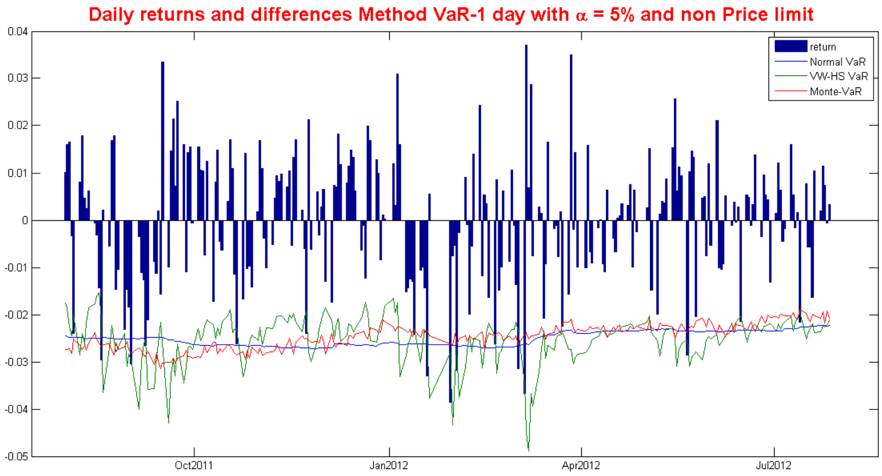

Figure 3: VaR Strategies Difference 2

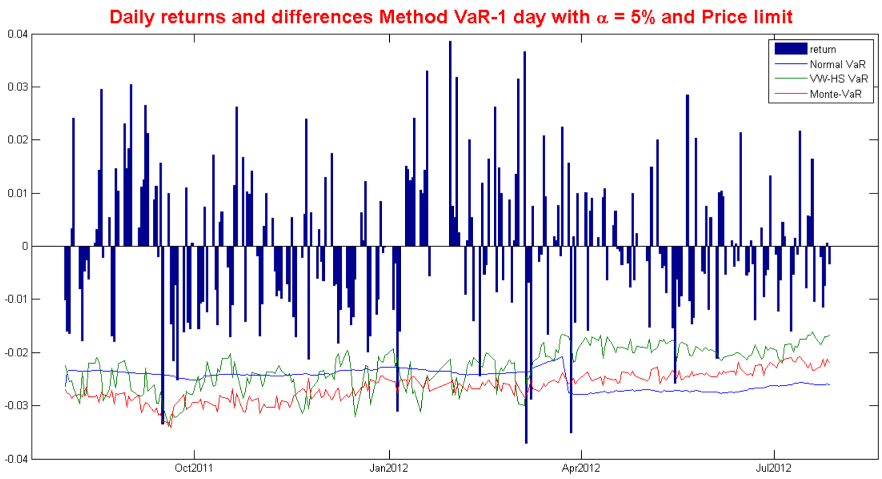

Figure 4: VaR Strategies Difference 3

6 Conclusion and Expansion

This report is our effort to represent some basic knowledges about the Value at risk concept and make it easy to practice in finance. This includes the mathematical understanding of VaR concept and the framework of calcu-

lating VaR methods in which the most important part is statistics of asset value. But statistics also is the weakness of every methods. The importance of assets distribution give us an intention to come back and make a better research about statistical problems. At last, it is also unforgettable to include testing for VaR or everything will become inefficient.
Beside all off these, there are some expansions in using VaR, such as portfolio optimization in [AG] or derivative hedging in [HP] and some more that any one can find around. Let review some examples.
In one portfolio optimization problem, we assume that the investor wants to invest in the market index and uses options to enhance the expected return of his portfolio (may be at a desire level) and at the same time fulfill the VaR constraint (or minimize that Value at Risk). The Value at Risk constraint is at the 99% level and the investor is not allowed to loose more than 10% of the initial portfolio value over the next ten days at this probability level.
In one hedging problem, we assume that one wants to hedge an option. But instead of using an initial capital that equals the price of the option, he accepts a shortfall probability of 5% that he can fail to hedge (or loss) to reduce the initial capital to a lower level.
In the last of this report, we will describe an allocation problem using VaR. Remind that VaR is defined as the largest amount of predict loss of a portfolio at a confident level (ex. 95%), (corresponding with 5% level of ruin) in a horizon of time (ex. one day).

$$P\left(L_{t,t+h} > VaR_{t,t+h}\left(\alpha\right)\right) = \alpha$$

Where P is probability measure $L_{t,t+h}$ is loss in time interval $(t, t+h)$, α is level of risk. One may consider to allocate his initial capital of x into two part, reserve and investment, that

$$x = R_0 + V_0$$

At time t his portfolio valued V_t and his reserve R_t should satisfy

$$P\left(V_{t+h} + R_t\left(\alpha\right) < 0\right) = \alpha$$

The required capital to cover the risk, or VaR also includes the current value of the portfolio:

$$VaR_{t,t+h}\left(\alpha\right) = R_t\left(\alpha\right) + V_t$$

That lead to

$$P\left(V_{t+h} - V_t < -VaR_{t,t+h}\left(\alpha\right)\right) = \alpha$$

or equivalent to

$$P\left(L_{t,t+h} > VaR_{t,t+h}\left(\alpha\right)\right) = \alpha$$

For example, we assume that the investor want to buy n stock with price S_0, that mean

$$V_0 = nS_0$$

He make that

$$P\left(nS_h + R_0\left(\alpha\right) < 0\right) = \alpha$$

or

$$P\left(n\left(S_h - S_0\right) + R_0\left(\alpha\right) + nS_0 < 0\right) = \alpha$$

that mean

$$x = R_0\left(\alpha\right) + nS_0 = nVaR_S\left(\alpha\right)$$

so,

$$n = \frac{x}{VaR_S\left(\alpha\right)}$$

Below are other problems that may be new in case of more complex allocation. We start with initial capital

$$x = R_0 + V_0$$

But, this time we want to allocate R_0 into a fixed amount and a variable amount that will be invested to a kind of portfolio P_t which has maturity T, that mean

$$R_0 = FR_0 + \theta P_0$$

Specially, the new portfolio P_t may be exactly the old portfolio V_t or simply, a bond, that

$$dP_t = \rho_t P_t$$

The very first problem is to minimize the fixed reserve FR_0 that equivalent to maximize the investing coefficient θ so that the VaR constraint is satisfied or

$$sup\left\{\theta > 0 : P\left(V_T + FR_0\left(\alpha\right) + \theta P_T < 0\right) = \alpha\right\}$$

And the main problem is maximizing the final total value in a set of admissible controls $\mathcal{A}$ of the process θ_t or

$$\sup_{\theta_t \in \mathcal{A}} \left(V_T + FR_T\left(\alpha\right) + \theta_T P_T\right)$$

under the conditions that

$$P\left(V_{t+h} + FR_t\left(\alpha\right) + \theta_t P_{t+h} < 0\right) \leq \alpha, t \in \left[0, T\right], h > 0.$$

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